

SURV/SURVMETH 720
Total Survey Error & Data Quality 1

Nonresponse error

Class #4
September 22, 2025

Methods for reducing coverage error

1. Redefine the target population

2. Enhance the sample frame

- a. Multiple frames

- b. Linking methods

- c. Network sampling

3. Statistical adjustments

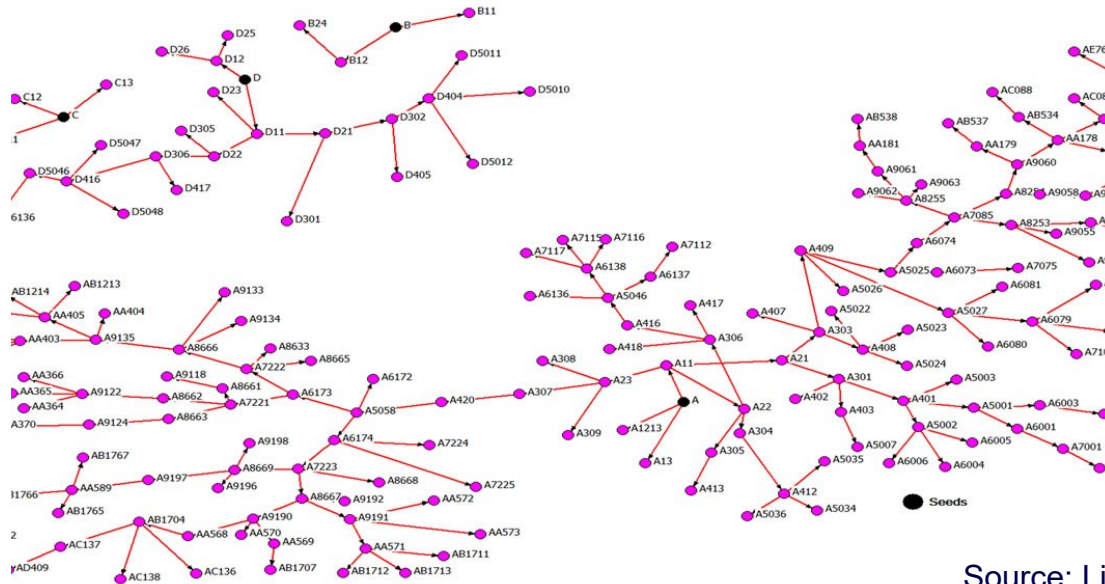
Hard-to-reach populations

Network Sampling

- What do we mean by “hard-to-reach”?
- Multiplicity
 - Respondents report about events among those in their personal networks (i.e., acquaintances, friends, extended family)
- Snowball sampling (non-probability)
 - Initial recruits in convenience sample refer to other members in the population.

Respondent-driven sampling (RDS)

- Effort to convert snowball sampling into a probability sampling method
- Recruit 'seeds' who then recruit others in their network



Source: Liu, Li, Ha, and Li (2012)

Some conditions for RDS

- Must be possible to trace who recruited whom
- Should collect information concerning the size of each person's network;
- Must have a maximum number of possible recruitments for a person
- Recruiter and recruit must have known each other prior to recruitment, each being a member of the other's social network.
- But not fully straightforward...

RDS – Potential Problems

- Can make strong assumptions about random selection, e.g., choice of network member is random.
- Recruitment...

...see Marpsata and Razafindratsima (2010).

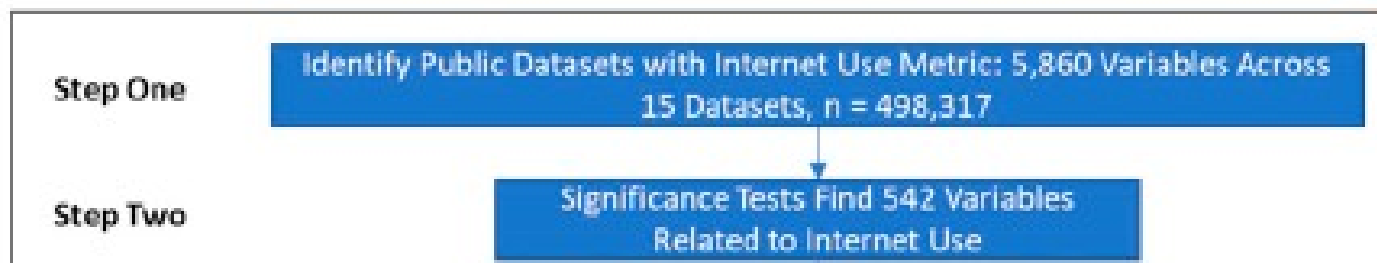
Methods for reducing coverage error

1. Redefine the target population
2. Enhance the sample frame
3. Statistical adjustments

Statistical adjustments

- General idea: Develop weights/calibrate to account for the non-covered population
- Example: Internet surveys leave out members with no internet access. Can weighting help bridge possible differences?

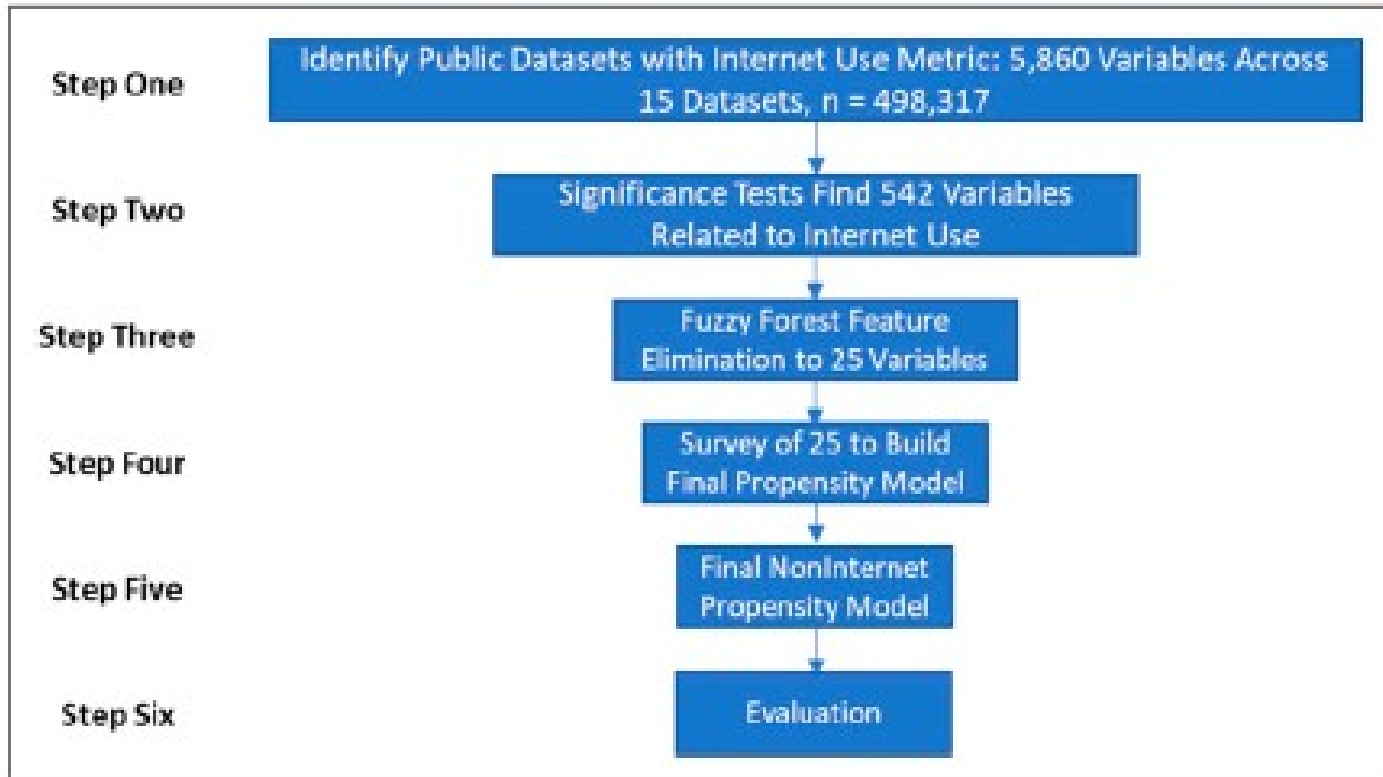
....Reducing Coverage Bias in Internet Surveys (Dutwin and Buskirk 2023)



- High internet penetration does not mean coverage bias is negligible.

authors reported their results. There are 122 variables for which the estimate for the internet population is at least double that for the non-internet population, or vice versa. To name just a few: own a computer (82% of the internet population vs. 20% for the non-internet population); went to see a movie in the past year (75% vs. 32%); own stocks (71% vs. 28%); became a U.S. citizen before 1960 (4% vs. 36%); importance to choose products politically (not important = 5% vs. 25%); and has given to charity (77% vs. 42%). Given these wide gaps, even with the non-internet population under 10 percent of the total population, a survey with only Internet respondents would experience bias of as much as 9% on these measures, and 105 variables would exhibit a bias of at least 5 percent.

....Reducing Coverage Bias in Internet Surveys (Dutwin and Buskirk 2023)



....Reducing Coverage Bias in Internet Surveys (Dutwin and Buskirk 2023)

Table 5. Random Forest Importance Measures, Final Omnibus Dataset.

Variable
Age
Go to movies past year
Know nanotechnology
Employment
Marital status
Education
Income
Know...Had a Baby...Year
Number of adults
Contacted politician...year

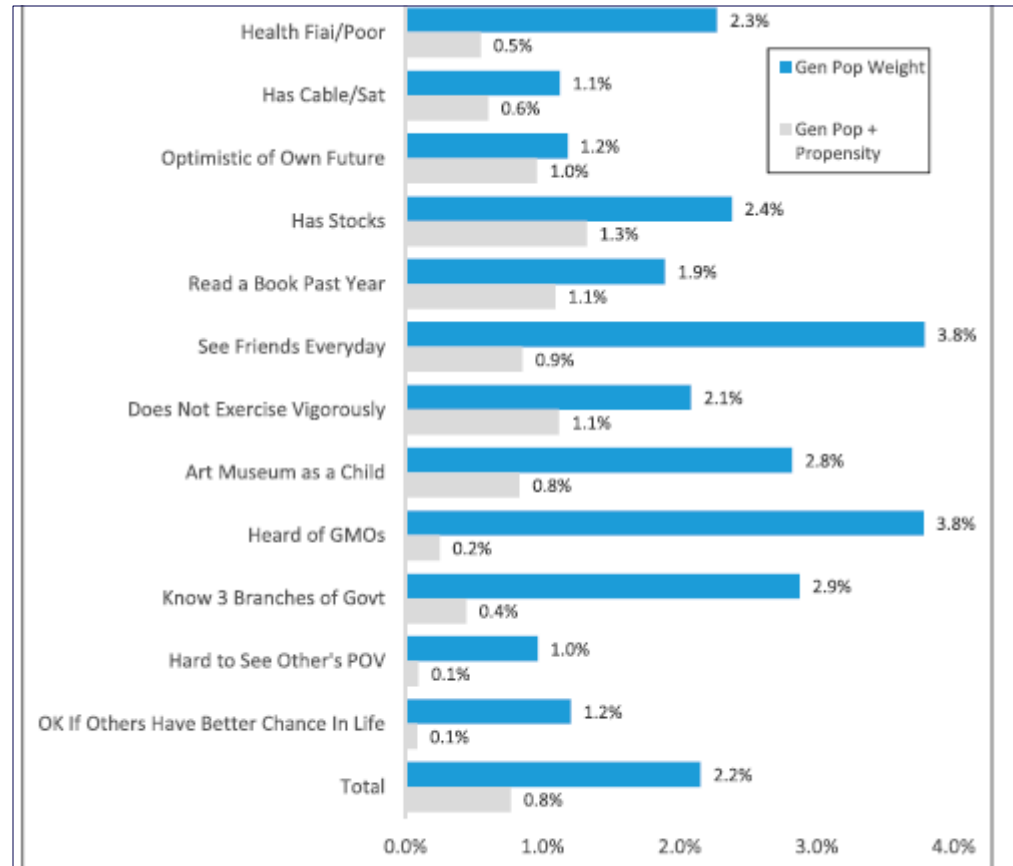


Figure 3. Mean absolute biases: Internet population, gen pop wgted versus gen pop + propensity wgted.

Also see Dever et al. (2008) for an assessment of weighting to reduce coverage bias

Consider a Web survey of the general population that omits those without Internet access. Those with access to the Internet differ from those without access on the survey variables. Do you see this as a "coverage" issue or "nonresponse" issue?

- Coverage
- Nonresponse

Providing Internet-Enabled Devices

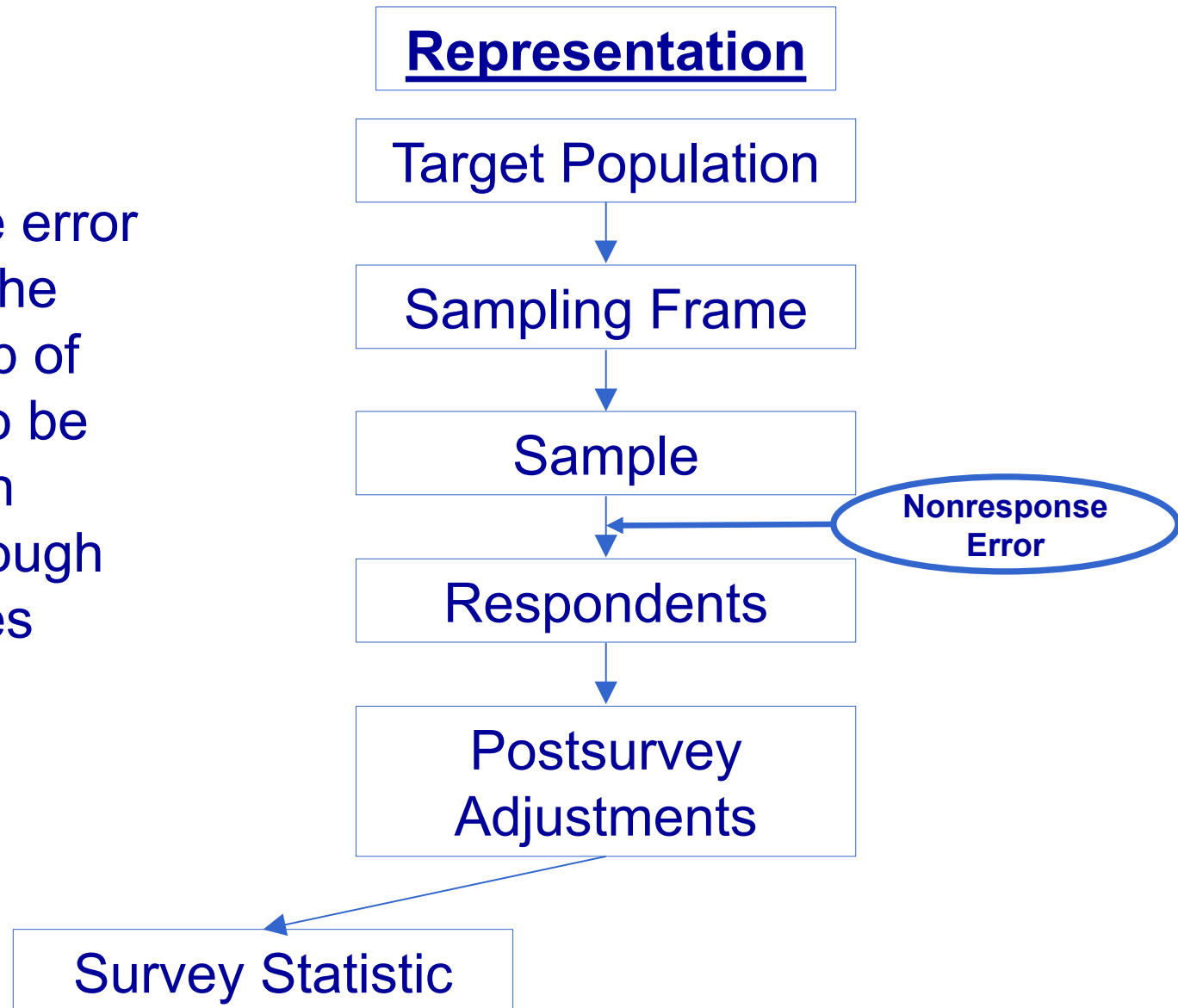
Examples

- LISS (Longitudinal Internet Studies for Social Sciences)
 - Selected households from Dutch population registry
 - Provided computer to non-Internet households
 - Eckman (2016): Exclusion of non-internet households in analyses of LISS data would bias a large number of estimates of means (but few of the estimates of regression coefficients)

Today's Agenda

- What causes nonresponse error
- How to measure nonresponse error

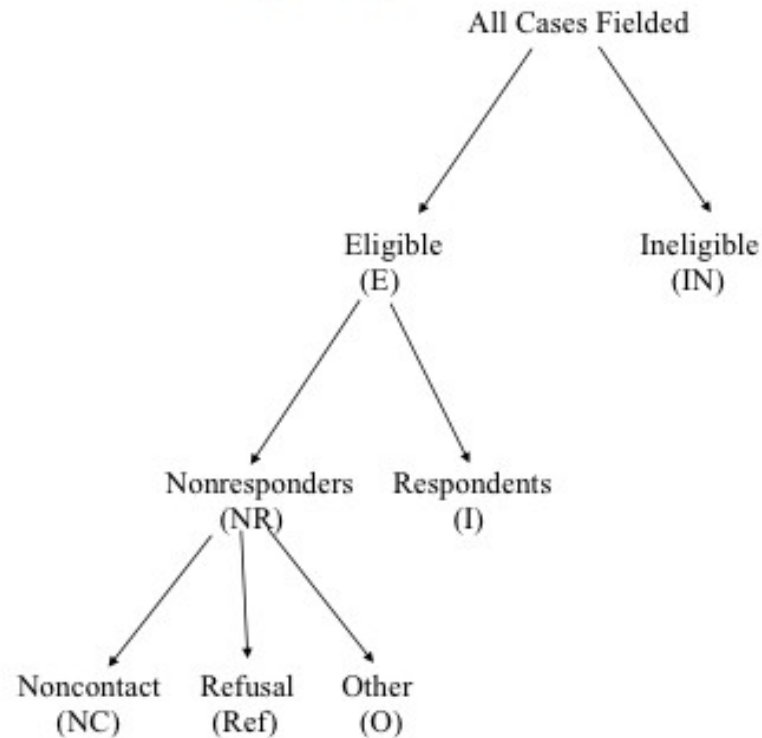
Nonresponse error occurs after the sampling step of survey (not to be confused with coverage, though consequences similar)



Nonresponse Error

- Arises when some sampled units do not respond
 - Unit nonresponse is the failure to obtain survey measures on a sample unit (our focus).
 - Item nonresponse is the failure to obtain an answer to a given item.

Components of Nonresponse



Refusals generally responsible for largest share of nonresponse in household surveys (but depends on other factors: geography, demography, survey, etc.)

Response Rates

- Proportion of estimated eligible sample who completed the survey
- Widely-used metric of survey participation:
 - Fairly simple to compute
 - Computation can be relatively standardized
 - Single number for the entire survey

AAPOR Dispositions for Response Rate Calculations

I = number of completed cases

P = number of partial cases

NC = number of noncontacted sample units known to be eligible

R = number of eligible refused cases

O = number of eligible sample units not responding for reasons other than refusal

U = number of sample units of unknown eligibility, not completed

e = estimated proportion of sample units of unknown eligibility that are eligible

AAPOR Response Rates

- $$\text{RR1} = \frac{I}{(I + P) + (NC + R + O) + U}$$
$$\text{RR2} = \frac{I + P}{(I + P) + (NC + R + O) + U}$$
$$\text{RR3} = \frac{I}{(I + P) + (NC + R + O) + eU}$$
$$\text{RR4} = \frac{I + P}{(I + P) + (NC + R + O) + eU}$$

Subjective Parts of Computation

- The minimum set of questions required for a “completed” interview
 - Criteria should be clearly specified
- Estimating share of cases with unknown eligibility that would have been eligible
 - Biemer, P., Murphy, J., & Kott, P. (2016). [Estimating Mail or Web Survey Eligibility for Undeliverable Addresses: A Latent Class Analysis Approach](#). In *JSM Proceedings* (pp. 1166-1172). American Statistical Association.

Other metrics : Contact rates, cooperation rates, and refusal rates

Nonresponse Rates over Time

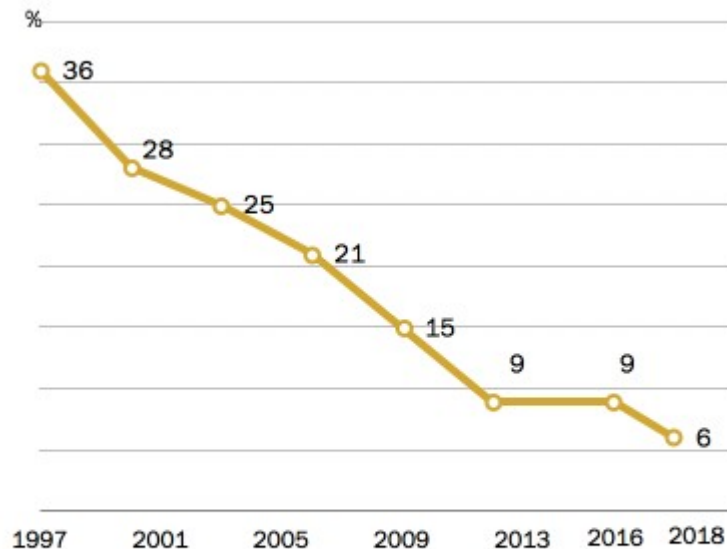
Overall trend of increasing nonresponse in household surveys

- Seems to be more pronounced in telephone surveys than in-person surveys

Non-response trends for telephone surveys...

From Pew Research

Response rate by year (%)



Note: Response rate is AAPOR RR3. Only landlines sampled 1997-2006. Rates are typical for surveys conducted in each year.

Source: Pew Research Center telephone surveys conducted 1997-2018.

AAPOR task force: 9 “large and prestigious” research firms

	Landline	Cell
2008	15.7%	11.7%
2009	13.7%	10.3%
2010	13.0%	11.2%
2011	13.6%	10.4%
2012	10.9%	7.2%
2013	9.8%	6.9%
2014	8.2%	6.5%
2015	9.3%	7.0%

<https://www.pewresearch.org/short-reads/2019/02/27/response-rates-in-telephone-surveys-have-resumed-their-decline/>

<https://aapor.org/wp-content/uploads/2022/11/Future-of-Telephone-Survey-Research-Report.pdf>

Survey	Year(s)	Overall Response Rates
^a California Health Interview Survey (CHIS)	2021–22	9.2%
^b National Health Interview Survey	2023	48.8%
^c Am Time Use Survey	2023	36.9%
BRFSS ^{d,e}	2023	44.7%

^d BRFSS response rates are presented here as median rates for all participating states and territories.

^e Response rate (RR) formulae may differ by survey:

CHIS—AAPOR RR#4 (like BRFSS)

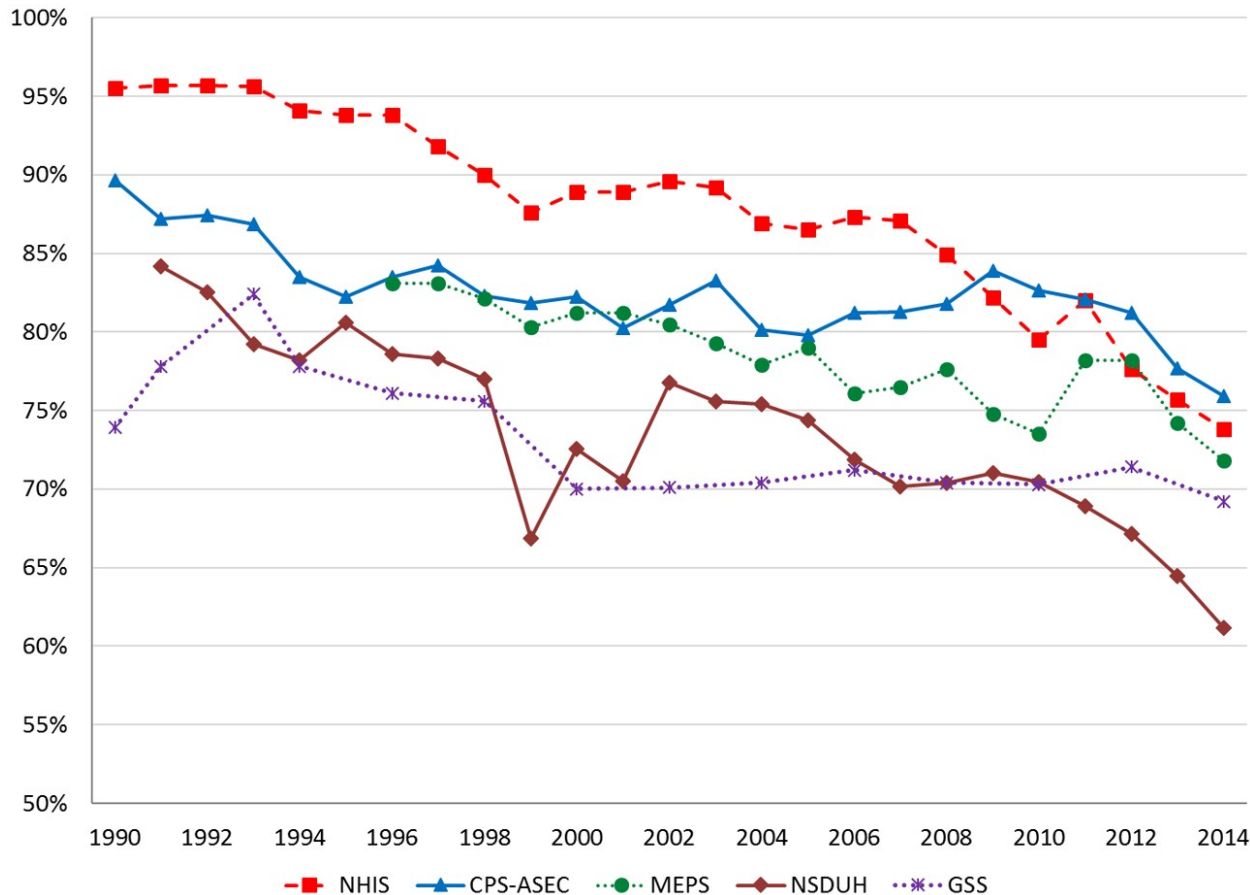
NHIS—AAPOR RR#2

ATUS—AAPOR RR#2

BRFSS, wide variation across states

State*	Landline Response Rate	Cell Phone Response Rate	Combined Response Rate
Minimum	30.4	17.1	21.7
Maximum	71.0	67.1	63.1
Mean	53.2	40.5	44.9
Median	54.3	40.3	44.6

In-person surveys...



NHIS: National Health Interview Survey

CPS-ASEC: Current Population Survey—Annual Social and Economic Supplement

MEPS: Medical Expenditure Panel Survey

NSDUH: National Survey on Drug Use and Health

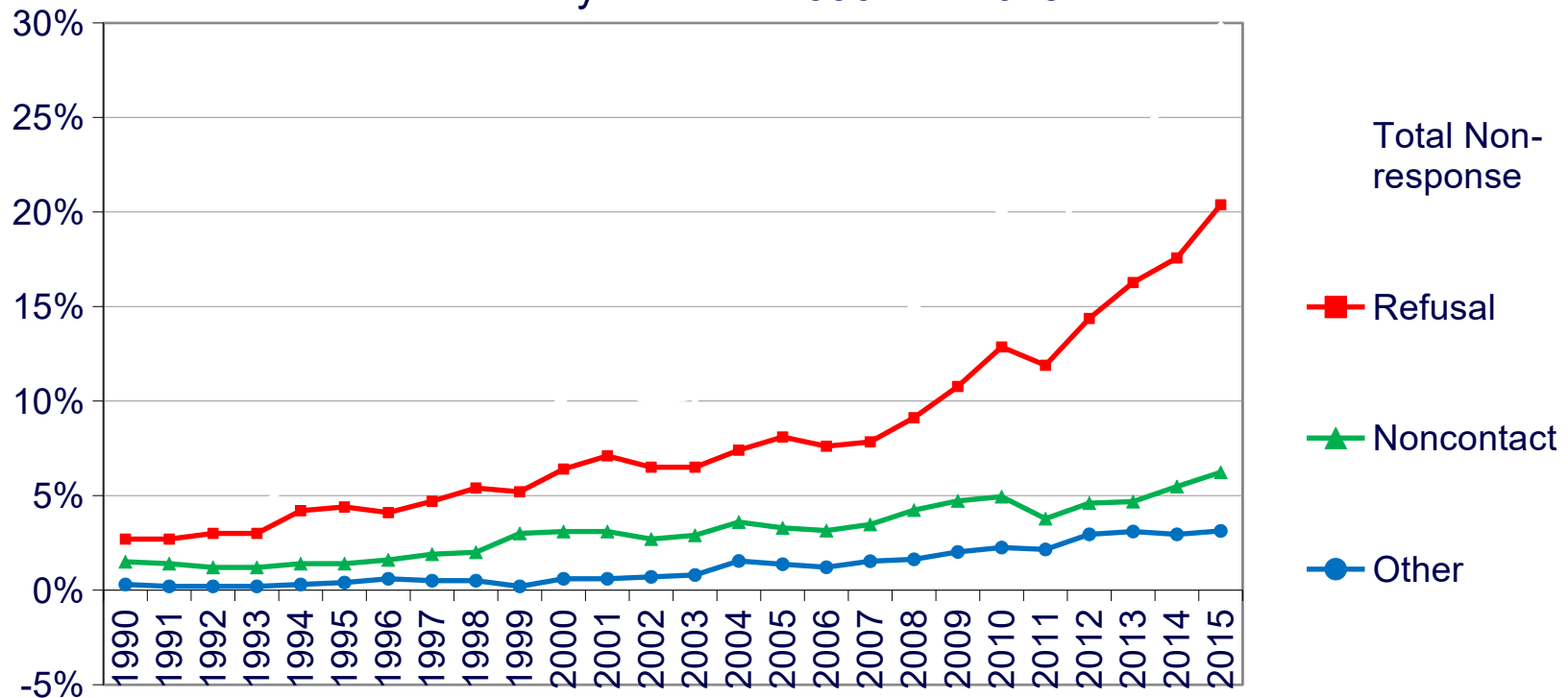
GSS: General Social Survey

Historical Response Rates

Source: Williams and Brick (2018)

Looking at contributions to non-response...

Unweighted Household Nonresponse in the National Health Interview Survey between 1990 and 2015



Grace, Brittany, Kehinde, Angelina: what about international surveys ?

Doris: Should developing national statistical offices continue aiming for $\geq 90\%$ response when U.S. evidence shows that high quality post-survey adjustments can keep bias small even with much lower response rates?

- Sagnik brings up the point about availability of auxiliary data in developing countries

International trends

- Luiten et al. (2020)
 - Response rates from labor force surveys from 1998-2015 in 25 countries (Western Europe and U.S.)
 - Average increase in nonresponse rate per year: 0.59 percent points
 - Equal contribution to decline by noncontacts and refusals
- India Human Development Survey (IHDS):
 - In-person component RR1: ~70% (2022-2024) down from 78% 12 years ago.
 - Telephone RR1: 33% despite doing an advance call.

Luz, Grace, Johnia, Lingchen, Felix,
Kehinde, Jeremy on “What would be ‘good’
response rates?”

What is a good response rate?

reduce nonresponse and encourage an adequate response rate.” Babbie (2007, p. 262) is bold enough to say, “A review of the published social research literature suggests that a response rate of at least 50 percent is considered adequate for analysis and reporting. A response of 60 percent is good; a response rate of 70 percent is very good.” Finally, Singleton and Straits (2005, p. 145) note, “Therefore, it is very important to pay attention to response rates. For interview surveys, a response rate of 85 percent is minimally adequate; below 70 percent there is a serious chance of bias.” All of these quotations come from books used to teach students about survey methods.

From Groves 2006

We do not recommend a use of *ad hoc* formulations of an ‘admissible extent’ of nonresponse in a survey. Rules or advice are sometimes proposed, stating that a minimal response rate, say 80 %, must be ascertained in a survey. Not only may the adherence to such rules prove costly from a data collection point of view, it can also result in a far from optimal use of the limited resources. The harmful effects of nonresponse, notably the bias in the estimates, can vary considerably from one survey to another. Adherence to fixed rules appears to be futile. An important question instead is to evaluate whether or not the survey results are likely to be severely distorted by the nonresponse. If they appear to be so, appropriate action is required.

Sarndal and Lundstorm, 2004, p.3



follow-up), sample design (e.g., coverage, frame), or other design issues. Any nonresponse adjustments should consider patterns of nonresponse.

Current OMB guidelines require a nonresponse bias study for all surveys with less than an 80% response rate; however, recent research indicates that nonresponse bias can occur in surveys with higher response rates and may not occur in studies with lower response rates. Plan for a nonresponse bias analysis if the expected unit response rates is below 80% (guideline 1.3.4) Plan for a nonresponse bias analysis if the expected item response rate is below 70% for any items used in reports (guideline 1.3.5).

Please note that OMB will generally not approve surveys with unit response rates below 60%. Therefore, in cases where prior experience suggests the potential for an overall unit response rate of less than 60%, the decision to proceed with data collection must be made in consultation with NCSES's chief statistician.

Federal Committee on Statistical Methodology (FCSM)

The FCSM Nonresponse Bias Subcommittee was chartered by the FCSM in 2016, ten years after the publication of the Office of Management and Budget (OMB) [Standards and Guidelines for Statistical Surveys](#). The OMB Standards and Guidelines for Statistical Surveys require all Federal surveys with a unit response rate of less than 80 percent to conduct an analysis of nonresponse bias (NRB). Since 2006, US Federal agencies, contractors, and data users have produced hundreds of NRB reports. However, the analytical and reporting practices for NRB have varied widely within and across agencies. The Subcommittee was chartered to advance the availability, comparability and quality of NRB analyses.

<https://www.fcsm.gov/resources/non-response-bias/>



**A Systematic Review of Nonresponse Bias Studies in
Federally Sponsored Surveys**

FCSM-20-02

[https://www.fcsm.gov/assets/files/docs/
A_Systematic_Review_of_Nonresponse_Bias_Studies_Federally_Sponsored_SurveysF
CSM_20_02_032920.pdf](https://www.fcsm.gov/assets/files/docs/A_Systematic_Review_of_Nonresponse_Bias_Studies_Federally_Sponsored_SurveysFCSM_20_02_032920.pdf)

...and so

- As a result, costs per completed interview have risen
- For example:
 - Cell-RDD “yield” (number of sample units required to secure a single completed interview) across 9 firms increased from 17 phone numbers per IW to 36 phone numbers per IW) (AAPOR task force report)
 - Cost per household for Census increased from \$16 in 1970 to \$94 in 2010 (in constant 2010 dollars) (GAO report 2016)

Potential Reasons for Nonresponse...

1. Barriers

- Explanation for increasing noncontact rates
- Relevant barrier depends on survey mode:
 - Face-to-face surveys: gated communities, locked apartment buildings
 - Telephone surveys: answering machines, Caller-ID, cell phones
 - Web surveys: spam filters

2. Busyness

- If people are busier, they may be hard-to-reach and reluctant to give up time
- Supporting this claim:
 - Increased labor force participation
 - Longer commute times
- Evidence against this claim
 - Abraham et al. (2006) find little evidence for impact of busyness (full-time vs. part-time vs. not working; hours worked; spouse's hours) on response rates in American Time Use Survey (ATUS)

Other reasons:

- people have less trust in survey organizations
- Survey fatigue (Eggleson 2024)
- Inability to provide data
- Yao, Kehinde, Zhenjing: sponsor effects, cultural norms
- Jay, Brittany: cognitive interviews/focus groups to figure out hurdles and what works

Three selected social science
theories...

1. Social Exchange Theory

(Homans 1961; developed for surveys by Dillman 1978)

- People respond if perceived benefits of responding outweigh costs of responding.
- Example benefits: having voice be heard; monetary incentive. Example costs: time, effort, loss of privacy.

Practically:

1. Provide small rewards through up-front incentives
2. Personalization (enhance importance)
3. Offering opportunity to contribute societally
4. Express appreciation for considering the request
5. Making the questionnaire appear interesting through the use of graphically designed covers.
(for mail surveys)

Greenman and Dillman (JSSAM 2023)

2. Social integration (isolation) theory (Goyder 1987)

- Individuals who are more integrated in society will be more likely to respond to a survey.
- Evidence for this claim:
 - Non-response $\leftrightarrow$ decreased civic participation such as voting and volunteering
 - Groves and Couper (1998): find lower contact rates and cooperation rates for those who live alone
 - Abraham et al. (2006): Lower response rates for those: living alone, not married, no school age children in household, rent rather than own

Social integration...

- Amaya and Harring (2017):
 - Look at different dimensions of integration: civic, family, neighborhood
 - With the ATUS, they find that civic engagement important but others not as much

Practically: Emphasize societal contribution in survey participation.

3. Leverage-Salience Theory

(Groves et al. 2000)

- People more likely to respond if features of the study that they like (have positive “leverage”) are emphasized (made “salient”) in the survey request.
- Calls attention to the need to tailor survey requests.
- Closely linked to a relatively new theory called pre-suasion (Cialdini 2016): events happening prior to making a specific request for a behavior can have a powerful influence on behavior.

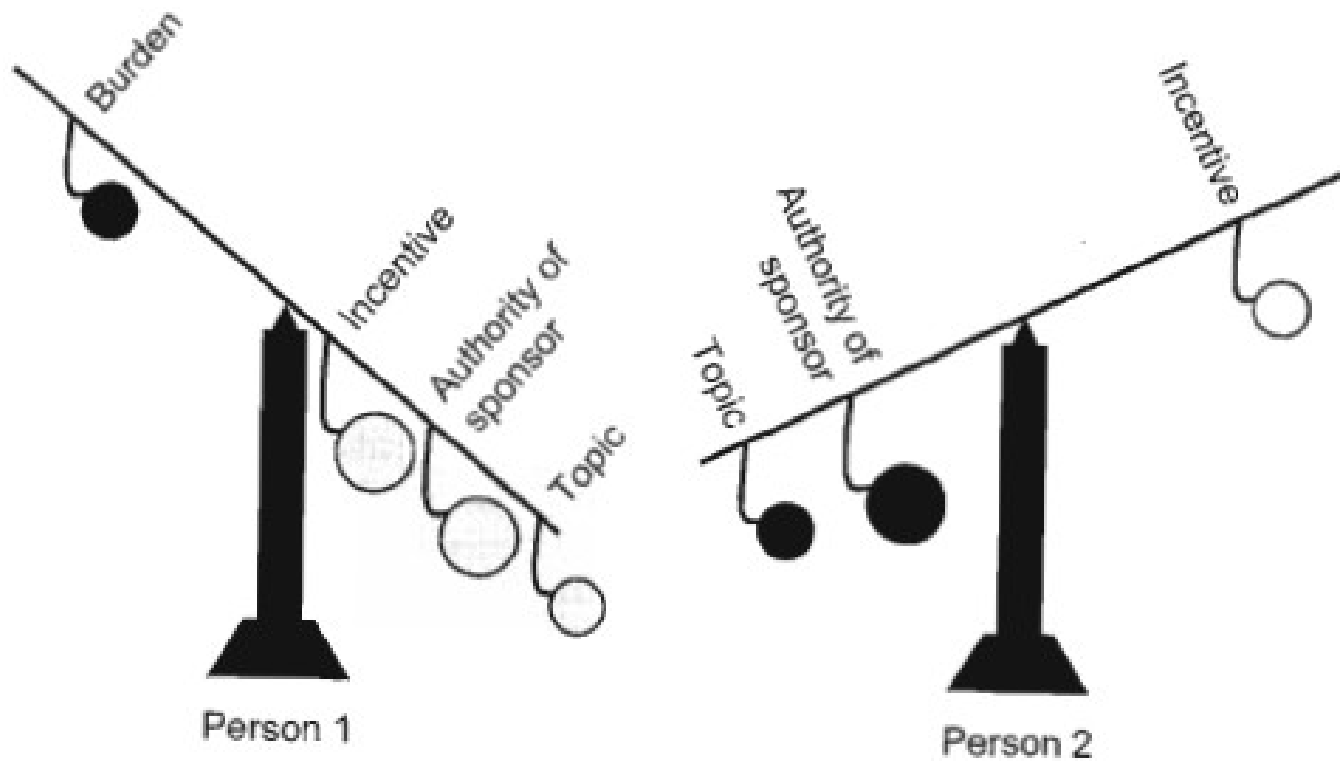


Figure 6.4 Two sample persons with different leverages for attributes of a survey request.

← Refusal Acceptance →

Limitation of Past Theories

- Focus on cooperation; do not address contact process.
- Assume people make rational choices based on features of the study; do not address the use of quick judgments/rules.
For example:
 - Throw out all mail, unless it's clearly a bill.
 - Hang up/don't answer telephone, unless it's a personal call from someone you know.

Measuring Nonresponse Error

Nonresponse Error

- Nonresponse bias in a respondent mean can be expressed as:

$$\text{Bias}(\bar{y}_r) = \bar{Y}_r - \bar{Y}$$

- $\bar{Y}$ is the mean of the full target population
- $\bar{Y}_r$ is the mean of respondents in the target population



..or frame population ?

Nonresponse Bias Continued

- Nonresponse bias in a respondent mean can also be expressed as:

$$\text{Bias}(\bar{y}_r) = \frac{M}{N} (\bar{Y}_r - \bar{Y}_m)$$

- Bias depends on the nonresponse rate and the difference between respondent and nonrespondent means in the target population

Views on Nonresponse

- **Deterministic view** (last formula): sample members either never respond or always respond conditional on the survey protocol.
- **Stochastic view:** *Propensity* to respond (ρ_i)

Xiaoqing: why have this stochastic view at all ?

Stochastic View

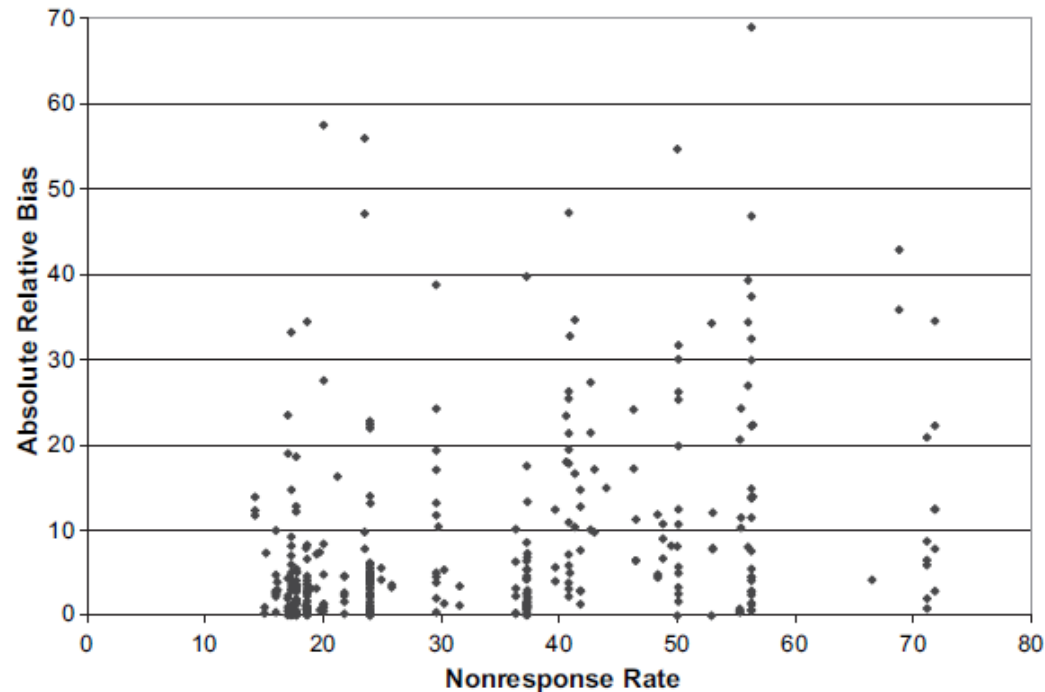
- Nonresponse bias for the respondent mean :

$$\text{Bias}(\bar{y}_r) = \frac{\sigma_{YP}}{\bar{P}}$$

- Depends on the covariance between P (response propensity) and Y (substantive variable) in the target population

Ted: How is this calculated in practice ?

From Groves (2006)



Ted: Would this still hold ?

Yujing: Would this hold across survey modes ?

Luz on “non-response rates are not necessarily bad *per se*”

Internationally...

“Do low survey response rates bias results? Evidence from Japan”, Rindfuss et al 2015

RESULTS

Evidence of response-rate bias is found for the univariate distributions of some demographic characteristics, behaviors, and attitudinal items. But when examining relationships between variables in a multivariate analysis, controlling for a variety of background variables, for most dependent variables we do not find evidence of bias from low response rates.

So, do you think you should prioritize achieving a high response rate?

Feiran, Zhenjing connect this to prevailing industry wisdom

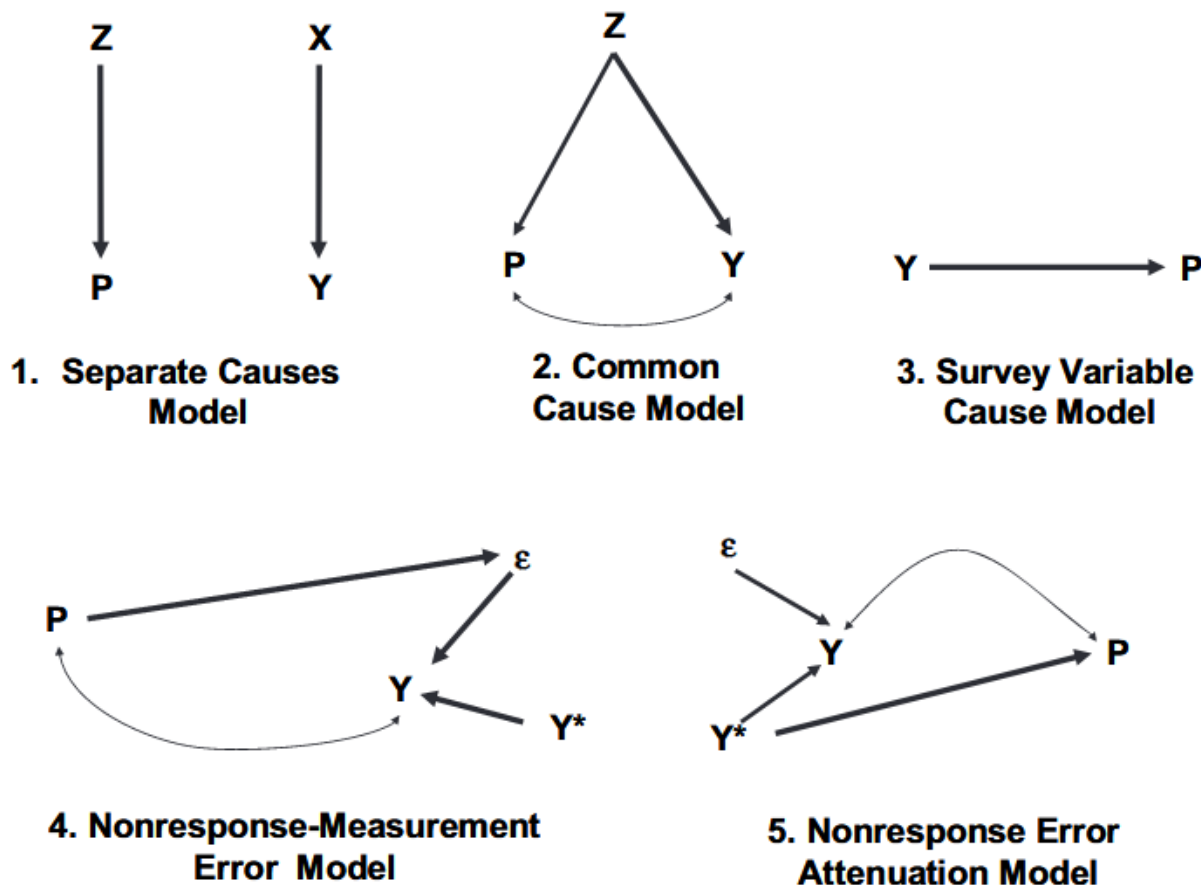


Figure 1. Five idealized causal models of response propensity (P), the reported survey variable (Y), the true value of the survey variable (Y^*) and other variables (X, Z) having different implications for nonresponse bias of the unadjusted respondent mean on the reported survey variable (Y).

Methods for Assessing Nonresponse Bias

1. Benchmarking

- Compare estimates obtained to estimates from other sources that are assumed to be more accurate
- Example: Keeter et al. (2000) compared estimates from “standard” 5-day survey (RR: 37%) to those from “rigorous” 8-week survey (RR: 61%)
 - Mean difference over 91 items around 2%
- Limitation: may confound NR error with other error sources

2. Compare Characteristics of Responders and Nonresponders

- Data about nonrespondents can come from:
 - Information on sampling frame
 - Paradata such as interviewer observations
 - Responses to screener interviews
 - Responses to other survey waves
 - Administrative records
- Example: Groves and Couper (1998)
 - Response to six household surveys evaluated using matched 1990 Census data
 - Lower contact rates for multiunit structures, single person households, households in urban areas.
 - Cooperation rates lower (refusal rates higher) in middle-aged households, those who live alone.

3. Compare Responses Given by Different Groups of Respondents

- Early vs. late responders
 - May be informative about the direction of response bias
- Nonresponse follow-up studies
 - Goal: to obtain responses from a random sample of the nonrespondents
- Those who express willingness to participate in a future survey vs. those who do not

Curtin et al (2000)

- Background: Constant RR (~70%) but increased effort (mean #calls: ~ 4 to 8) needed over the years.
- What happens if we exclude respondents who
 - Needed refusal conversion ? (RR reduced by 5-10 %)
 - Needed more than 5 calls ? (RR reduces 25%)
 - Needed more than 2 calls ? (RR reduces 50%)

Small to no effect on key estimate! (and on the component estimates)

But is this really about absolute nonresponse bias ?

Index of Consumer Sentiment (y)

- While small,
 - Initial cooperators to Refusal conversions
 - y for Initial cooperators $>$ Initial refusers
 - Early call IWs to later call IWs
 - y for early calls $<$ later calls

4. Compare y values on respondents and non-respondents (methodological studies)

Amaya and Presser (2017)

- Use frame information from the American Time Use Survey (ATUS) and the Survey of Health, Ageing, and Retirement in Europe (SHARE).
 - ATUS drawn from CPS.
 - SHARE is a panel survey.
- Compare **estimates** from the full sample to the respondent sample.
- H1: Social integration hypothesis. Those who are social will also respond to surveys. So estimates on social activities and roles should be biased upwards. [yes]
- H2: Variables measuring political and social activities should be more biased than those measuring other activities (e.g., hosting a dinner party) [some evidence]
- H3: Regression coefficients predicting these activities unimpacted [impact present but small].

Next Week's Class

- Dealing with the problem of nonresponse
- Questions?

Thank you !